

UEI626: PROSTHETICS AND REHABILITATION

L	T	P	Cr
3	0	0	3.0

Course Objectives: The objective of the course is to provide basic understanding of rehabilitation technology and assistive technologies for physically impaired.

Introduction: Introduction to types of physical impairments, Rehabilitation Engineering, Motor, Sensor and Communication Disorders and Introduction to Artificial Organs

Seating Mechanics and Systems: Introduction to seating systems and interventions required, Design aspects of different wheel chairs such as Manual wheelchairs, electric power wheelchairs, Wheel chair standards & tests

Prosthetic Devices: Anatomy of human body, Pervasive assistive technologies, Computer assistive technologies, Cognitive assistive technologies, Artificial limb and hands, intelligent prosthetics, externally powered and controlled orthotics and prosthetics, Advanced hybrid assistive system, Myoelectric hand and arm prostheses, The MARCUS intelligent hand prostheses

Vision and Hearing Aid: Anatomy of eye and ear, Classes of ear and visual impairments and assistive technologies, Implants and aids for sight and hearing impaired

Advanced Rehabilitation Robotics: A case study of design of advanced robots for rehabilitation of physically and mentally impaired, Neural engineering in rehabilitation engineering.

Course Learning Outcomes (CLO):

Student will be able to:

1. Demonstrate the knowledge of assistive technologies for rehabilitation engineering (SDG 3)
2. Establish the understanding of working principle and applications of seating mechanics and wheel chair designs (SDG 11)
3. Select a prosthetic device for physically impaired
4. Exhibit understanding of the anatomy of different organs and implants and aids for respective impairments

Text Books:

1. Rory A Cooper, Hisaichi Ohnabe, Douglas A Hodson: *An Introduction to Rehabilitation Engineering*, CRC Press, First edition, 2006
2. Marion A Hersh, Michael A Johnson, *Assistive Technology for Visually impaired and blind people*; Springer Publications, First edition, 2008
3. Suzanne Robitaille, “*The illustrated guide to Assistive technology and devices–Tools and gadgets for living independently*”, Demos Health Newyork, First edition, 2010.

Evaluation Scheme:

Sr. No.	Evaluation Elements	Weightage (%)
1	MST+EST	70
2	Sessional (May include Assignments//Quizzes/Lab Evaluations)	30

Approved in 109th meeting of the Senate held on March 16, 2023. Revised in 112th and 114th meeting of the Senate held on March 11, 2024 and March 7, 2025 respectively.